

PG&E Climate Solutions

by Charles Segerstrom

In July 2007, Pacific Gas & Electric Company (PG&E) hosted the Affordable Comfort, Incorporated, (ACI) summit “Moving Existing Homes Toward Carbon Neutrality.” The summit brought together 99 motivated experts, mostly from the United States and Canada, to wrestle with the issues of achieving deep energy reductions in our housing stock. This two-day summit was the first bold step toward the important long-term goal of reducing total energy consumption of North American homes, individually and collectively, to a small fraction of what it is today. Achieving this will require nothing short of a paradigm shift in our approach to energy efficiency.

In Germany—a country committed to the Kyoto Protocol—scientists and policymakers recognize that, based on Earth’s estimated carbon-carrying capacity of 10 billion tons per year and a global human population that will reach ten billion by 2050, we need to limit our carbon emissions to 1 ton per person per year. In Germany and Europe, current carbon emissions are about 10 tons per person per year, so bringing them down to a sustainable level requires a reduction by a factor of ten. In the United States, however, carbon emissions are now about 20 tons per person per year—with California lower than the national average—so we will need to work twice as hard to reach the 1-ton goal.

What are Americans willing to do to reduce our residential carbon emissions to sustainable levels? Summit participants realize that the business-as-usual approach must be replaced with new thinking. For example, instead of aiming to incrementally improve air conditioners and lower duct leakage, we need to seal and insulate homes to the point that we can



To work toward reducing GHGs, PG&E is aggressively adding renewable electric power resources to its supply of delivered power, as well as training building professionals to make homes as affordable, comfortable, and energy-efficient as possible.

eliminate forced-air systems altogether. We need new low-energy systems that heat both space and water, and that are appropriately sized to meet dramatically reduced loads. Instead of undervaluing home energy savings and performance improvements because our current cost-benefit models are overconstrained, we should develop new methods of valuing these improvements. Rather than short-term programs that target low-hanging fruit, we need programs whose long-term goals are deep energy reductions, with all intermediate measures in alignment with those goals.

The opportunities and challenges identified during the ACI summit were both enlightening and alarming. In the United States, buildings generate more greenhouse gas emissions than automobiles—39% and 21%, respectively, of the total, according to the Energy Information Administration—and existing homes are the largest contributor to the building sector’s carbon footprint. Retrofitting homes is much more difficult and expensive than constructing energy-efficient new homes. And even in California, with its progressive Title 24 building energy code, many new homes need to be improved as soon as they are occupied. Existing homes present a unique set of

challenges for reducing total energy consumption, because the occupants’ attitudes and behavior are critical to achieving deep home energy reductions.

So how do we begin to implement strategies for this new paradigm of deep residential energy reduction? Home performance programs across the United States are establishing new standards for home energy retrofits, based on a systematic building science approach to resolving comfort, indoor air quality, and energy problems. These programs deserve the ongoing support of federal and state

governments and utilities; they should become the cornerstone of deep energy retrofits and the basis of other efforts to “green” existing homes. Home performance advocates are calling for the development of a green-collar workforce to meet the demand for the labor needed to mitigate carbon emissions. ACI will continue to explore and develop the potential for achieving deep home energy reductions in the United States and Canada, and PG&E will maintain a leading role in reducing greenhouse gas emissions from California and U.S. homes.

Convinced that climate change is a real and urgent problem, PG&E’s CEO, Peter Darbee, testified before Congress in 2007 to urge the federal government to respond; he is a key supporter of Governor Schwarzenegger’s cap and trade policy, and was recognized by the Natural Resources Defense Council (NRDC) as being a “tireless advocate for long overdue mandatory national limits on global warming pollution.” PG&E’s Energy Training Center—where I have worked for 26 years—pioneered diagnostic testing of homes in California. We held our first blower door classes and combustion safety training in 1991 and our first duct leakage testing classes in 1993, and we sponsored Affordable Comfort’s first West Coast conference in 1994.

Our residential classes now include building science topics and training for home performance contractors, and many qualify for continuing education credits for members of the National Association of the Remodeling Industry (NARI) and Build It Green professionals. PG&E will continue to take a proactive role in training building professionals to make homes as affordable, comfortable, and energy-efficient as possible.

Much of the effort needed to achieve deep energy reduction in homes will happen outside the regulatory sphere, but other states can learn from California's progressive energy policy, which decouples energy sales from utility profits to remove disincentives for energy efficiency. California also has a loading order for future energy supplies, requiring that energy efficiency be implemented before renewables, demand response mechanisms, or building new conventional power plants. It is no accident that the ACI summit was held in California, and PG&E looks forward to working with ACI, Home Energy, and others to take the next steps necessary to demonstrate deep home energy reductions—and to enable us all to achieve them.



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